Day 1: Introduction to Networking and the OSI ModelTheory

Tasks 1 : Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.

To capture network traffic using Wireshark and identify the protocols used during internet communication.

Procedure

1. Open Wireshark on the computer.
2. Select the active network interface, such as Wi-Fi.
3. Start capturing packets.
4. Use the internet while Wireshark is capturing traffic.
5. Stop the capture and observe the **Protocol** and **Info** columns.
6. Identify the protocols shown in the captured packets.

Protocols Observed

Protocol	Purpose
TCP	Provides reliable data delivery and uses acknowledgements to confirm received data.
UDP	Transfers data without establishing a TCP connection.
ARP	Helps find the MAC address associated with an IPv4 address on the local network.
TLSv1.2	Helps encrypt data exchanged during secure communication.
IPv6	Provides addressing and routing for communication across networks.

Observation

The capture contains TCP, UDP, ARP and TLSv1.2 packets. An ARP request asks which device has IP address 192.168.1.6, and the reply provides its MAC address. Several TCP packets use port 443, and TLSv1.2 packets show application data being exchanged.

Task 2 : Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc.).

To identify different OSI layers by examining network packets captured in Wireshark.

Procedure

1. Open the captured traffic file in Wireshark.
2. Observe the **Protocol** and **Info** columns in the packet list.
3. Select three packets with different protocols.
4. Identify the OSI layer mainly associated with each protocol.
5. Record the packet number, protocol, layer, and its function.

Packet Analysis

Packet No.	Protocol	OSI Layer	Function
1	TCP	Layer 4 – Transport	TCP provides reliable communication between devices. It uses sequence numbers and acknowledgements to help ensure data is delivered correctly.
7	ARP	Layer 2 – Data Link (commonly mapped)	ARP helps a device find the MAC address associated with an IPv4 address on the local network.
13	TLSv1.2	Layer 6 – Presentation (traditional mapping)	TLS encrypts data to help protect communication between a client and a server.

Observations

1. Packet 1 – TCP

The Protocol column shows TCP, and the Info column shows an ACK (acknowledgement). TCP is associated with the Transport Layer because it supports reliable data delivery and acknowledgement of received data.

2. Packet 7 – ARP

The packet contains the message: “Who has 192.168.1.6? Tell 192.168.1.1”. This is an ARP request asking which device uses that IPv4 address. ARP is commonly associated with the Data Link Layer in basic networking studies.

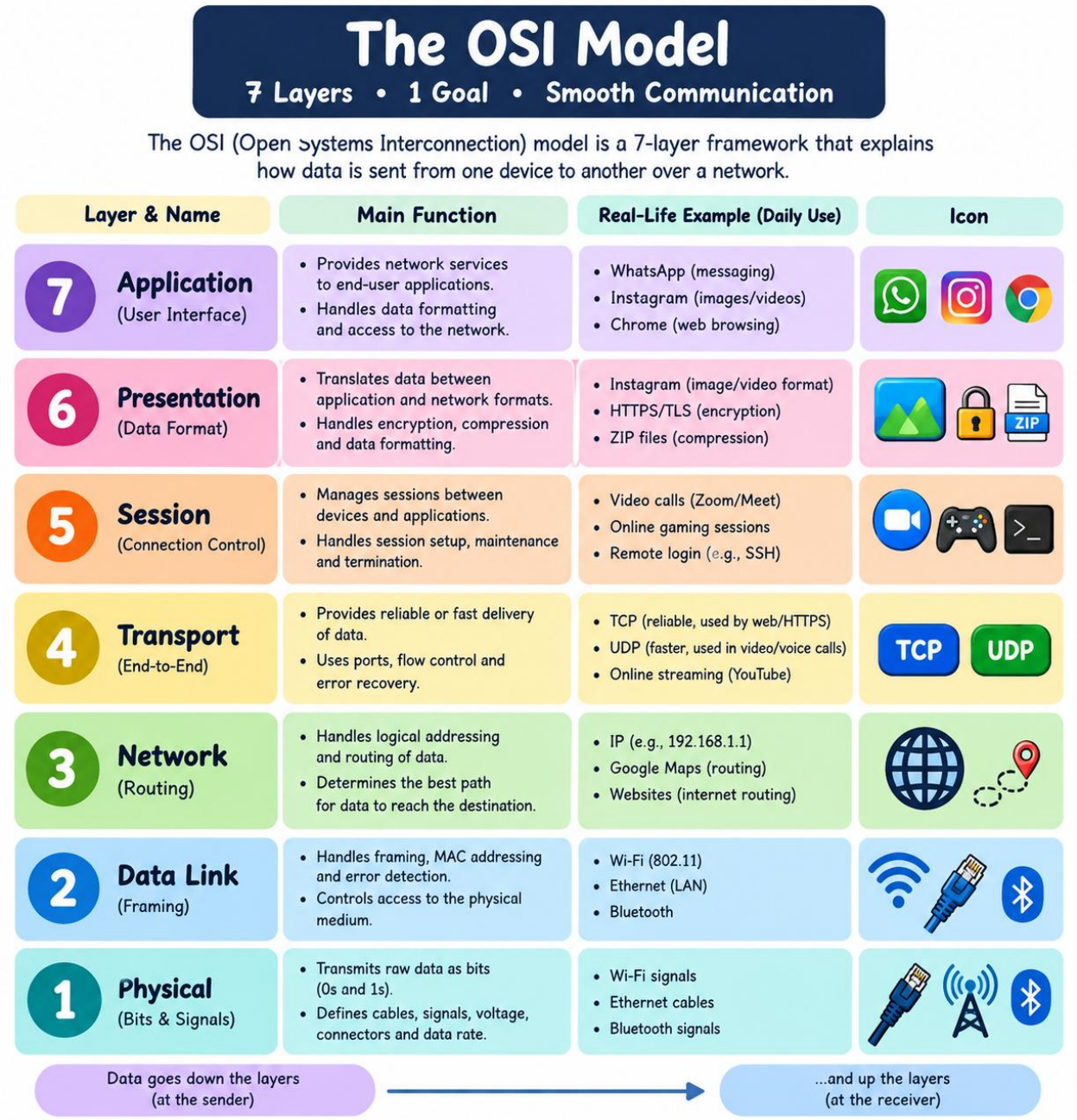
3. Packet 13 – TLSv1.2

The Protocol column shows TLSv1.2, and the Info column shows “Application Data”. TLS protects data by encrypting it during communication. In the traditional OSI teaching model, encryption is associated with the Presentation Layer.

Conclusion

Wireshark helped identify protocols used in network communication and relate them to the OSI model. TCP is associated with the Transport Layer, ARP with the Data Link Layer, and TLS with the Presentation Layer in the traditional mapping. These observations demonstrate how different networking functions work together during communication.

Task 3 : Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).



Task 4 : Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words.

To understand how a WhatsApp message travels from the sender's device to the receiver's device using networking layers.

Working Process

Suppose I send **“Hello”** to my friend on WhatsApp. The message passes through different networking layers before reaching the receiver.

1. **Layer 7 – Application Layer:** WhatsApp allows me to type and send the message.
2. **Layer 6 – Presentation Layer:** Data is represented in a suitable format and protected using encryption. WhatsApp uses end-to-end encryption for personal messages.
3. **Layer 5 – Session Layer:** Communication between devices is maintained. In real networks, session-related functions may be handled across different protocols and layers.
4. **Layer 4 – Transport Layer:** Transport protocols help deliver data between applications. TCP or other transport mechanisms may be used depending on the connection.
5. **Layer 3 – Network Layer:** IP addressing helps route data across networks toward the destination.
6. **Layer 2 – Data Link Layer:** Data is placed into frames and sent across the local network, such as Wi-Fi.
7. **Layer 1 – Physical Layer:** The data travels as signals through Wi-Fi radio waves, cables, or other physical media.

At the Receiver's End

When the message reaches my friend's device, the data is processed through the networking layers in reverse order. The device processes the received data, and WhatsApp displays **“Hello”** on the screen.